sed

Overview

- sed -> stream editor
- used to edit files
- general form:

```
sed [options] '/pattern/ command'
input_file
sed [options] 'range command'
input_file
```

Helpful Options

- -E -> use extended regular expression
- -f program_filename -> read sed commands from file named program_filename
- --in-place [=suffix] -> modify file in place, optionally give suffix to create backup file with suffix extension

Common Uses

- Deleting lines:
  - sed -E '/^Totals.*/ d'
  - sed '4 d'
  - sed '1, 4 d'
- $ in range form =end of file

Substituting Text

- Substituting Text:
  - sed -E 's/pattern_to_match/replace_with/'
  - sed -E 's/ *$//' filename
- default: only replaces 1st occurrence of pattern on each line

Substituting Text (continued)

- can specify to replace others:
  - replace all occurrences
    - sed -E 's/pattern_to_match/replace_with/g' fname.txt
  - replace certain occurrence
    - sed -E 's/pattern_to_match/replace_with/2' fname.txt

Substituting Text (continued)

- combine line range with substitution
  - sed '1, 3 s/pattern_to_match/replace_with/' fname.txt
- use matched pattern (or subexpression) in replace_with
  - \1,\2,... refer to sub-expressions grouped in parentheses
  - & -> entire pattern matched